Trader Behavior vs Market Sentiment

Project report — ds_report.pdf

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1. Executive summary

This analysis examines how trader behavior (profitability, leverage, trade volume, and side) aligns with Bitcoin market sentiment — categorized as **Fear** or **Greed**. Using historical trade records from Hyperliquid (per-trade level) merged with a Fear & Greed index for corresponding dates, we quantified differences in performance and risk between sentiment regimes.

Key findings (high-level):

- **Leverage increases on average during Greed periods**, suggesting traders take on more risk when market sentiment is optimistic.

- **Aggregate PnL** is not strictly higher during Greed; at times, **contrarian profits** (trades in Fear) outperform, indicating opportunity during panic.

- **Profitability %** differs by side (buy/sell) and sentiment; buy trades tend to dominate during Greed, while profitable sell trades can increase in Fear.

- **Large leverage with Greed** correlates with higher PnL variance — higher upside and greater downside risk.

2. Data

Trader dataset (Hyperliquid) — per-trade fields used:

- `account`, `symbol`, `execution price`, `size`, `side`, `time`, `start position`, `event`, `closedPnL`, `leverage`

Sentiment dataset (Fear & Greed):

- `Date`, `Classification` (values: `Fear`, `Greed`)

Merged dataset: `trader_data` left-joined to `sentiment` on `Trade Date` → `Date`.

3. Methodology

3.1 Data preparation (Power Query)

- Convert `time` (timestamp) to DateTime.

- Create `Trade Date` (date-only) as key for merging with sentiment.

- Filter invalid trades (zero size, null closedPnL).

- Merge with sentiment index (left join) and expand `Classification`.

3.2 Key calculated columns & measures

- **TradeDirection**: `if side = "buy" then 1 else -1`

- **ProfitFlag**: `closedPnL > 0`

- DAX measures:

  - `Total PnL = SUM(trader_data[closedPnL])`

  - `Total Volume = SUM(trader_data[size])`

  - `Average Leverage = AVERAGE(trader_data[leverage])`

  - `Total Trades = COUNTROWS(trader_data)`

  - `Profitable Trades = CALCULATE(COUNTROWS(trader_data), trader_data[closedPnL] > 0)`

  - `Profitability % = DIVIDE([Profitable Trades],[Total Trades],0)`

(Exact DAX snippets are included in the appendix.)

3.3 Visuals

- KPI cards (Total PnL, Average Leverage, Total Trades, Profitability %)

- Profit by Sentiment (clustered column)

- Trade Volume Over Time (line chart)

- Leverage vs PnL (scatter)

- Profitability by Side (stacked column)

- Sentiment Timeline (area)

4. Results & interpretation

4.1 Leverage behaviour

- **Observation**: Average leverage increases during Greed windows.

- **Interpretation**: Traders appear more willing to use borrowed capital when market sentiment is bullish/greedy; this raises systemic risk in those windows.

4.2 Profitability and sentiment

- **Observation**: Although volume and leverage are higher in Greed, **mean closedPnL** is not always greater than in Fear.

- **Interpretation**: Higher leverage magnifies outcomes — while some traders capture outsized profits, many also record larger losses. Fear periods can present contrarian profitability opportunities (lower volume, less leverage, but clearer mean reversion patterns).

4.3 Side (buy/sell) differences

- **Observation**: Buy trades represent a greater share of volume & profits in Greed; sell trades increase in prevalence and occasionally in profitability during Fear.

- **Interpretation**: This matches behavioral finance expectations where investor optimism drives longer position sizes.

4.4 Risk vs Reward (scatter)

- **Observation**: The scatter of `closedPnL` vs `leverage` shows higher variance at higher leverage, particularly in Greed.

- **Interpretation**: Risk management is especially important during Greed windows.

5. Example actionable insights

1. **Dynamic leverage caps** — lower maximum allowable leverage during strong Greed signals to reduce tail risk exposure.

2. **Sentiment-aware sizing** — scale trade size down during high-leverage/Greed periods unless supported by additional signals.

3. **Contrarian strategy filter** — monitor Fear windows for mean-reversion signals: smaller size, lower leverage, and selective entries could yield favorable risk-adjusted returns.

4. **Alerts** — set automated alerts when Average Leverage and Sentiment both spike.

6. Limitations

- Sentiment index is daily and coarse; intraday sentiment fluctuations within a day are not captured.

- Trader-level metadata (experience, balance) not available — some accounts may consistently use different risk profiles.

- The merge uses `Trade Date` (date-only), so timestamp-level sentiment mismatches (intraday sentiment) are not resolved.

- Survivorship bias: bad trades might be missing if only closed positions are recorded.

7. Next steps / future work

- Add intraday sentiment signals (if available) or social media / on-chain indicators for higher granularity.

- Segment traders by account-level metrics (AUM, historical win rate) to test heterogeneity.

- Backtest simple rules (e.g., leverage cap during Greed) and evaluate PnL and drawdown changes.

- Incorporate transaction costs and slippage to more accurately estimate realized PnL.
